MILITARY SPECIFICATION

CROMOLYN SODIUM CAPSULES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Cromolyn Sodium Capsules (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Box, Folding, Paperboard.
PPP-B-676	Box, Setup.
PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals. (See 6.3)

STANDARDS

Federal

FED-STD-00285	Capsules (for Medicinal Purposes).

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN PHARMACEUTICAL ASSOCIATION PUBLICATION

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042).

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, D. C. 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Cromolyn Sodium Capsules, and shall be in accordance with all applicable requirements of FED-STD-00285, and as specified herein:

S2 Classification. Shall be Type I, Class 1.

S5.4 The following additional requirements and tests are added to this paragraph:

Each capsule shall contain 20 mg of Cromolyn Sodium within the designated assay limits for the capsule. The capsules shall assay to contain not less than 100.0 percent and not more than 122.0 percent of the required amount of Cromolyn Sodium when assayed as specified in 4.3.1.

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Shall be suitable for use with 6515-00-133-5426, Inhaler, Medicament, Plastic.

Identity. The active ingredients in the finished capsules shall comply with the identity tests specified in 4.3.2.

Content Uniformity. Shall comply with the content uniformity test as specified in 4.3.3.

Microbial Limits. Shall contain not more than 100 microorganisms per gram with no Staphylococcus aureus or Pseudomonos aeruginosa when determined in accordance with the USP method or similar method yielding equivalent results.

Inhalation (Emptying) Test. Capsules shall have an average weight loss of not less than 90 percent with no more than 5 percent of the samples having a weight loss of less than 75 percent when tested as specified in 4.3.4.

Weight Variation. Shall comply with the requirements specified in 4.3.5.

Paragraph S6.2.9 of FED-STD-00285 shall not apply.

S6.1 Ingredients. The following additional requirements are added to this paragraph:

The Cromolyn Sodium used in the manufacture of the finished capsules shall comply with the following:

Identity. Shall comply with identity tests specified in 4.4.1.

Free acidity or Free Alkalinity. Not more than 0.25 ml of 0.1N Sodium Hydroxide or not more than 0.25 ml of 0.1N Hydrochloric Acid shall be required to make a solution containing 1.0 gram of Cromolyn Sodium dissolved in 20 ml of boiled purified water neutral to bromothymol blue.

Arsenic. Shall contain not more than 2 PPM Arsenic when determined using the U.S.P. method.

Lead. Shall contain not more than 10 PPM Lead when determined using the U.S.P. method.

Sulfated Ash. Shall be not less than 26.6 percent and not more than 28.6 percent calculated with reference to dried material, when tested as specified in 4.4.2.

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Loss on Drying. Shall lose not more than 10.0 percent of its weight on drying when tested as specified in 4.4.3.

Assay. Shall assay not less than 97.0 percent and not less than 101.5 percent Cromolyn Sodium when tested as specified in 4.4.4.

All other ingredients used in the manufacture of the finished capsules shall comply with S5.2.

3.2 Potency period. Shall have an expiration dating period (potency period) of 36 months.

4. REQUIREMENTS

4.1 Supplier Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. The classification of defects as shown in FED-STD-00285 shall be applicable.

4.3 Tests

4.3.1 Assay of Cromolyn Sodium in Filled Capsule.

Reagents. Buffer Solution. Dissolve 70 grams of Sodium Phosphate, Dibasic, Anhydrous in 900 ml of purified water. Adjust to pH 7.4 by adding 10 percent phosphoric acid. Dilute to 1000 ml with purified water and mix well. Dilute 100 ml of the above solution to 1000 ml with purified water and mix well. (This final solution must be freshly prepared.)

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Standard solution. Weigh accurately about 80 mg of Cromolyn Sodium reference standard and transfer to a 100 ml volumetric flask. Dissolve in and dilute to volume with purified water and mix well. Pipet a 10 ml aliquot into a 250 ml volumetric flask, add 1 ml of buffer solution and dilute to volume with purified water. At the same time, determine the moisture content of another portion of the Cromolyn Sodium reference standard by drying at 105°C for 16 hours over P_2O_5.

Procedure.

Accurately weigh one finished capsule, and carefully transfer the contents to a 25 ml volumetric flask. Reweigh the empty gelatin capsule immediately, avoiding handling it any more than is necessary. Dissolve the contents of the flask and dilute to the mark with purified water. Transfer a 5.0 ml aliquot of this solution to a 200 ml volumetric flask. Add 1 ml of buffer solution and dilute to the mark with purified water. Concomitantly determine the absorbances of both solutions in 1 cm cells at a wavelength of maximum absorbance at approximately 326 nm, with a suitable spectrophotometer using a 1:200 w/v buffer solution as the blank.

Calculate the mg Cromolyn Sodium per capsule as follows:

$$\frac{Au}{As} \times \frac{Wt\,Std}{100} \times \frac{(100-n)}{100} \times \frac{10}{250} \times \frac{25}{5} \times 200$$

where Au = absorbance of sample
As = absorbance of standard
Wt (std) = Weight of standard in mg
n = percent moisture in standard

4.3.2 Identity. The ultraviolet absorption of the sample solution used in the assay shall exhibit maxima at 238 nm and 326 nm.

4.3.3 Content Uniformity. Shall comply with the U.S.P. Content Uniformity Test for Capsules, using the assay method specified in 4.4.1, with a target fill of 22.2 mg cromolyn sodium per capsule.

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4.3.4 Inhalation (Emptying) Test.

Apparatus. An apparatus constructed in such a way that one or more plastic Medicament Inhalers (6515-00-133-5426) may be attached by their mouthpieces to an apparatus having an aspirational action. The volume of air aspirated through each loaded inhaler (containing one capsule) shall be at the rate of 60 liters per minute per inhaler.

Procedure

Weigh not less than 20 filled capsules (W_1). Place each capsule in respective cups of the plastic Medicament Inhalers. Pierce the capsules making sure that two holes have been formed in each of the capsules. Fit the loaded inhalers into the apparatus. Subject each capsule to four aspiration cycles within 10 seconds (at the rate of 60 liters/minute). Remove the capsules from the inhalers. Wipe off any powder on the outside of the capsules and reweigh the capsules (W_2). Open the capsules, remove the remaining powder and reweigh the empty capsules (W_3). Calculate the fill weight (W_1-W_3), the weight emptied (W_1-W_2) and hence the percentage emptied for each capsule. Calculate the average percentage (by weight) of powder expelled and the number of capsules less than 75% emptied. If the capsules fail the requirements, an additional 40 capsules shall be tested. The average of the sixty percentages emptied shall be not less than 90% by weight, and not more than 3 of the 60 capsules shall have less than 75% by weight of its powder expelled.

4.3.5 Weight Variation

Weigh the contents of 20 capsules individually and determine the average net weight. Not more than two of the individual net weights shall be less than 87.5 percent nor more than 112.5 percent of the average net weight and no net weight shall be less than 75 percent nor more than 125 percent of the average net weight. If more than 2 but less than 6 capsules deviate from the average net weight by between 12.5 percent and 25 percent, determine the net weight of an additional 40 capsules and determine the average net weight of the entire 60 capsules. Not more than six of the 60 individual net weights shall be less than 87.5 nor more than 112.5 percent of the average net weight and no capsule shall differ by more than 25 percent.

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4.4 Ingredients.

4.4.1 Identity Test.

4.4.1.1 Color reaction. Dissolve 10 mg p-nitro-aniline in 10 ml of purified water, cool to 0°-5°C, add 1 ml dilute hydrochloric acid, holding temperature below 5°C during the subsequent addition of 10 mg of sodium nitrite. Mix well. Dissolve 10 mg of cromolyn sodium in 5 ml purified water. Add 1 ml of 10 percent sodium hydroxide and boil gently for 1 minute Cool to below 5°C. Add 0.5 ml of p-nitro-aniline reagent and mix. A deep red color shall form.

4.4.1.2 Infrared absorption. The infrared absorption spectrum of a potassium bromide dispersion of it, previously dried at 105° C under vacuum overnight, shall exhibit maxima and minima at the same wavelengths as that of a similar preparation of Cromolyn Sodium reference standard.

4.4.1.3 Flame Test

Shall yield a strong, positive flame for sodium.

4.4.1.4 U. V. Absorption. The ultraviolet absorption spectrum of the assay solution shall exhibit maxima and minima at the same wavelengths as that of a similar preparation of Cromolyn Sodium reference standard.

4.4.2 Sulfated Ash

Weigh accurately 1.0 gram of the powdered sample in a tared silica crucible. Moisten with concentrated sulfuric acid. Ignite gently over a Meker burner, again moisten with sulfuric acid and re-ignite over a bunsen flame, cover crucible; heat strongly to constant weight.

4.4.3 Loss on Drying.

Weigh accurately about 2 gram of the cromolyn sodium into an Aluminum dish with tight fitting cover. Remove the cover and dry for 16 hours in a vacuum drying oven, containing a dish of phosphorus pentoxide, at 105°C and 5 mm Hg pressure. The oven should be bled with a current of dry air during drying to ensure removal of all volatiles produced.

Remove the container from the oven and replace cover, allow to cool in a desiccator and weigh.

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4.4.4 Assay

Reagents and Standard Solution: Refer to 4.4.1

Procedure

Accurately weigh about 400 mg of Cromolyn Sodium and transfer to a 250 ml volumetric flask, dissolve in and dilute to volume with purified water and mix well.

Pipet a 5.0 ml aliquot into a 250 ml volumetric flask, add 1 ml of buffer solution and dilute to volume with purified water. Mix well.

Concomitantly determine the absorbance of both the standard and sample solutions in 1 cm cells at a wavelength of maximum absorbance at approximately 326 nm, with a suitable spectrophotometer using a 1:200 v/v buffer solution as a blank.

4.5 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table I. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table I. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and Packing Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper location, sequence, or method of application.

NOTE: The above examination is only applicable to the unit, intermediate and exterior containers.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for Delivery shall be in accordance with all applicable paragraphs of PPP-C-186 and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One box containing sixty individually packaged capsules, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. Capsules shall be individually sealed in an immediate container which shall be acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. Sixty individually sealed capsules shall be packaged in a box of an appropriate size constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be adequate to prevent accidental opening.

5.2.2.3 Intermediate package. Twelve units packages shall be packaged as specified in 5.2.7.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. Twelve intermediate packages (144 units) shall be packed in an exterior container as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twelve intermediate packages (144 units) shall be packed in an exterior container as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

5.4.1.2 Unit package. Each unit package label shall bear the following:

(a) labeling information in accordance with commercial practice.
(b) the National Stock Number.
(c) the expiration date,

Note: (1) The National Stock Number shall not be proceeded by any prefix.
(2) Strip labeling may be used.
(3) Labeling adhesion test shall not apply.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life markings as specified in MIL-STD-129 shall be shown.

5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Marking shall include type I shelf life markings as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) the National Stock No.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

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6.2 This specification covers the following item:

NATIONAL STOCK NO.	ITEM IDENTIFICATION
6505-00-133-5421	CROMOLYN SODIUM CAPSULES, Individually Sealed, 20 mg per capsule, 60s

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superceded by a revision.

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1388
is

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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MIL-C-37325

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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